Grade 11 Practice

Standard N.CN.1

Name: _____ Date: _____

Computation

1. Solve: i^2 = _____ **-1**

2. Solve: i^3 = _____ **-i**

3. Solve: i^4 = _____ **1**

4. Write $\sqrt{-9}$ in terms of i. _____ **3i**

Word problems

5. Write $\sqrt{-16}$ as a complex number.

Show your work and answer.

Answer: 4i

6. What is the real part of 3 + 4i?

Show your work and answer.

Answer: 3

Grade 11 Practice

Standard N.CN.2

Name: _____ Date: _____

Computation

1. Solve: (2 + 3i) + (1 − i) = _____

3 + 2i

2. Solve: (1 + i)(1 − i) = _____

2

3. Solve: (2 + i)(2 − i) = _____

5

4. Solve: (3 + 2i) − (1 + 4i) = _____

2 − 2i

Word problems

5. An impedance is (4 + 2i) plus (1 + i). What is the total?

Show your work and answer.

Answer: 5 + 3i

6. Solve: i(3 + i) = _____

Show your work and answer.

Answer: −1 + 3i

Grade 11 Practice

Standard N.CN.7

Name: _____ Date: _____

Computation

1. Solve $x^2 + 1 = 0$. x = _____

±i

2. Solve $x^2 + 4 = 0$. x = _____

±2i

3. Solve $x^2 + 9 = 0$. x = _____

±3i

4. Solve $x^2 + 2 = 0$. x = _____

±i√2

Word problems

5. What is the sum of the roots of $x^2 + 1 = 0$?

Show your work and answer.

Answer: 0

6. What is the product of the roots of $x^2 + 4 = 0$?

Show your work and answer.

Answer: 4

Grade 11 Practice

Standard A.SSE.3

Name: _____ Date: _____

Computation

1. Write $x^2 + 6x + 5$ in vertex form. _____

(x + 3)² − 4

2. What is the minimum value of $x^2 - 4x + 7$? _____

3

3. What is the sum of the roots of $x^2 - 5x + 6 = 0$? _____

5

4. What is the product of the roots of $2x^2 + 8x + 6 = 0$? _____

3

Word problems

5. A ball's height is $h = -16t^2 + 32t$, where t is seconds. What is its maximum height?

Show your work and answer.

Answer: 16 feet

6. The function $y = -x^2 + 6x$ has a maximum value. What is it?

Show your work and answer.

Answer: 9

Grade 11 Practice

Standard A.APR.2

Name: _____ Date: _____

Computation

1. Let $p(x) = x^2 - 3x + 2$. What is $p(1)$? _____

0

2. Let $p(x) = x^3 - x$. What is $p(2)$? _____

6

3. Is $x - 2$ a factor of $x^2 - 4$? Write yes or no.

yes

4. What is the remainder when $x^2 + 1$ is divided by $x - 1$? _____

2

Word problems

5. Is $x + 1$ a factor of $x^3 + 1$? Write yes or no.

Show your work and answer.

Answer: yes

6. What is the remainder when x^3 is divided by $x - 2$?

Show your work and answer.

Answer: 8

Grade 11 Practice

Standard A.REI.2

Name: _____ Date: _____

Computation

1. Solve $2/x = 4$. $x =$ _____

$\frac{1}{2}$

2. Solve $6/(x + 1) = 3$. $x =$ _____

1

3. Solve $x/3 = 2/x$. Write both solutions.

$\pm\sqrt{6}$

4. Solve $(x + 2)/x = 3$. $x =$ _____

1

Word problems

5. A rate is $12/x = 3$. What is x?

Show your work and answer.

Answer: 4

6. A tank fills in $6/x$ hours, and the time is 2 hours. What is x?

Show your work and answer.

Answer: 3

Grade 11 Practice

Standard A.REI.7

Name: _____ Date: _____

Computation

1. Solve the system $y = x$ and $y = x^2$. Write the solutions.

(0, 0) and (1, 1)

2. Solve the system $y = 2$ and $y = x^2$. Write the x-values.

$\pm\sqrt{2}$

3. Solve the system $y = x + 2$ and $y = x^2$. Write the solutions.

(2, 4) and (-1, 1)

4. Solve the system $y = 0$ and $y = x^2 - 4$. Write the x-values.

±2

Word problems

5. A path $y = x^2$ reaches ground level $y = 0$. Where does it meet the ground?

Show your work and answer.

Answer: (0, 0)

6. A circle $x^2 + y^2 = 25$ meets the line $y = 0$. What are the x-values?

Show your work and answer.

Answer: ±5

Grade 11 Practice

Standard A.REI.11

Name: _____ Date: _____

Computation

1. Find where $y = 2x$ meets $y = x + 3$. Write (x, y).

(3, 6)

2. Find the x-values where $y = x^2$ meets $y = 4$.

±2

3. Find where $y = 3x - 1$ meets $y = x + 5$. Write (x, y).

(3, 8)

4. Find where $y = 1/x$ meets $y = x$. Write the solutions.

(1, 1) and (−1, −1)

Word problems

5. Costs $y = 10x$ and $y = 5x + 50$ are equal at what x?

Show your work and answer.

Answer: x = 10

6. A parabola $y = x^2 - 4$ meets the line $y = 0$. What are the x-values?

Show your work and answer.

Answer: ±2

Grade 11 Practice

Standard F.IF.6

Name: _____ Date: _____

Computation

1. Find the average rate of change of $f(x) = 2x + 3$ on $[1, 4]$. _____ **2**

2. Find the average rate of change of $f(x) = x^2$ on $[0, 2]$. _____ **2**

3. Find the average rate of change of $f(x) = x^2$ on $[1, 3]$. _____ **4**

4. Find the average rate of change of $f(x) = 5$ on $[0, 10]$. _____ **0**

Word problems

5. A distance is $d = 3t$, where t is hours. What is the average rate of change from $t = 0$ to $t = 4$?

Show your work and answer.

Answer: 3

6. A temperature is 20 at $t = 0$ and 40 at $t = 4$. What is the average rate of change per hour?

Show your work and answer.

Answer: 5

Grade 11 Practice

Standard F.IF.7

Name: _____ Date: _____

Computation

1. Where is the vertex of $y = |x|$? Write the ordered pair. **(0, 0)**

2. What is the y-intercept of $y = x^2 + 2$? _____ **2**

3. Where is the vertex of $y = -x^2$? Write the ordered pair. **(0, 0)**

4. What is the domain start of $y = \sqrt{x}$? _____ **0**

Word problems

5. Where is the vertex of $y = |x - 3|$? Write the x-value.

Show your work and answer.

Answer: 3

6. What is the y-intercept of $y = 2^x$?

Show your work and answer.

Answer: 1

Grade 11 Practice

Standard F.IF.8

Name: _____ Date: _____

Computation

1. Write $y = x^2 + 6x + 5$ in vertex form. _____

$(x + 3)^2 - 4$

2. Write y = 2^(3x) with base 8. _____

8^x

3. Write y = 1.05^(2t) with base 1.1025. _____

1.1025^t

4. Factor $y = x^2 - 4$. _____

$(x - 2)(x + 2)$

Word problems

5. A balance is y = 1000(1.05)^t after t years. What is the starting balance?

Show your work and answer.

Answer: **1000**

6. Write a factored form of a quadratic with roots 3 and -1.

Show your work and answer.

Answer: **$(x - 3)(x + 1)$**

Grade 11 Practice

Standard F.IF.9

Name: _____ Date: _____

Computation

1. Compare $f(x) = 2x + 1$ and $g(x) = 3x$. Which has the greater slope? Write f or g.

g

2. Which grows faster in the long run, a linear or an exponential function? Write your answer.

exponential

3. Which has the greater y-intercept, $f(x) = x^2 + 5$ or $g(x) = 2^x$? Write f or g.

f

4. Which function is bounded above, $f(x) = -x^2 + 4$ or $g(x) = 2x$? Write f or g.

f

Word problems

5. Plan A costs $30 + $5 per hour and Plan B costs $10 per hour. Which plan costs less at 8 hours? Write A or B.

Show your work and answer.

Answer: A

6. A ball has height $h = -16t^2 + 64t$. What is its maximum height?

Show your work and answer.

Answer: 64

Name: _____ Date: _____

Computation

1. The graph of $y = f(x) + 3$ is shifted how far, and in which direction? Write your answer.

up 3

2. The graph of $y = f(x - 2)$ is shifted in which direction? Write right or left.

right

3. The graph of $y = -f(x)$ is reflected over which axis? Write x or y.

x-axis

4. The graph of $y = f(2x)$ is compressed in which direction? Write horizontally or vertically.

horizontally

Word problems

5. The graph of $y = x^2$ is shifted left 4 units. Write the new equation.

Show your work and answer.

Answer: $y = (x + 4)^2$

6. The graph of $y = f(x)$ is stretched vertically by 3. Write the new equation.

Show your work and answer.

Answer: $y = 3f(x)$

Grade 11 Practice

Standard F.BF.4

Name: _____ Date: _____

Computation

1. Find the inverse of $f(x) = 2x + 6$. $f^{-1}(x) =$ _____

(x - 6)/2

2. Find the inverse of $f(x) = x^3$. $f^{-1}(x) =$ _____

x^($\frac{1}{3}$)

3. Find the inverse of $f(x) = x + 5$. $f^{-1}(x) =$ _____

x - 5

4. If $f(x) = 2x$, what is $f^{-1}(f(4))$? _____

4

Word problems

5. The formula $F = (\frac{9}{5})C + 32$ converts Celsius to Fahrenheit. Write the inverse formula for C.

Show your work and answer.

Answer: **$C = (\frac{5}{9})(F - 32)$**

6. If $f(x) = 3x - 1$, what is $f^{-1}(8)$?

Show your work and answer.

Answer: **3**

Grade 11 Practice

Standard F.LE.4

Name: _____ Date: _____

Computation

1. Solve $\log_2 8$ = _____ **3**

2. Solve $\log_{10} 1000$ = _____ **3**

3. Solve $\log_3 9$ = _____ **2**

4. Solve $\log_5 1$ = _____ **0**

Word problems

5. Solve $2^x = 32$ for x.

Show your work and answer.

Answer: 5

6. Solve $10^x = 100$ for x.

Show your work and answer.

Answer: 2

Grade 11 Practice

Standard F.TF.1

Name: _____ Date: _____

Computation

1. π radians equals how many degrees? _____ **180**

2. π/2 radians equals how many degrees? _____ **90**

3. 2π radians equals how many degrees? _____ **360**

4. Write 60° in radians. _____ **π/3**

Word problems

5. A wheel turns through 90°. Write the angle in radians.

Show your work and answer.

Answer: π/2

6. Write π/4 radians in degrees.

Show your work and answer.

Answer: 45°

Grade 11 Practice

Standard F.TF.2

Name: _____ Date: _____

Computation

1. Find cos 0. _____ **1**

2. Find sin(π/2). _____ **1**

3. Find cos π. _____ **-1**

4. Find sin 0. _____ **0**

Word problems

5. On the unit circle, what point is at angle π/2?

Show your work and answer.

Answer: (0, 1)

6. Find sin(π/6).

Show your work and answer.

Answer: $\frac{1}{2}$

Grade 11 Practice

Standard F.TF.5

Name: _____ Date: _____

Computation

1. What is the period of sin x? Write in terms of π.

2π

2. What is the amplitude of y = 3 sin x? _____

3

3. What is the period of y = sin(2x)? Write in terms of π.

π

4. What is the midline of y = 2 cos x + 5? Write the equation.

y = 5

Word problems

5. Daylight hours follow h = 12 + 3 sin(t). What is the amplitude?

Show your work and answer.

Answer: 3

6. A tide oscillates between 1 foot and 9 feet. What is its amplitude?

Show your work and answer.

Answer: 4

Name: _____ Date: _____

Computation

1. A scatter plot has a positive trend. Write positive or negative for the correlation.

positive

2. A line of best fit is y = 2x + 1. What is the predicted y when x = 4? _____

9

3. Does an outlier affect a regression line? Write yes or no.

yes

4. A residual is observed minus predicted. Write the word for the difference.

residual

Word problems

5. A model is y = 0.5x + 10. What is the prediction when x = 20?

Show your work and answer.

Answer: 20

6. A data point is 15 and the model predicts 12. What is the residual?

Show your work and answer.

Answer: 3

Grade 11 Practice

Standard S.ID.7

Name: _____ Date: _____

Computation

1. A line has slope 3. What does this mean for y as x increases by 1?

y increases by 3

2. In cost = 50 + 2x, what is the fixed cost? _____

50

3. A line has slope -1.5. How does y change when x increases by 1?

decreases by 1.5

4. In y = 40 + 5x, what does the intercept 40 represent? Write starting value or rate.

starting value

Word problems

5. Cost is 20 + 4x, where x is items. What does the slope represent?

Show your work and answer.

Answer: **$4 per item**

6. Temperature is T = 60 - 2h, where h is hours. What does the slope represent?

Show your work and answer.

Answer: **drops 2 degrees per hour**